

2023 Bi H1 Q24

Section: Sustainability and Interdependence

Topic: Components of Biodiversity

Question Summary

The graph shows how the populations of bush babies and spider monkeys changed between 2015 and 2019 in a tropical rainforest. You are asked which of the four statements is supported by the data.

Worked Solution

From the graph:

- In 2015, bush babies had a population of around 180, while spider monkeys were about 90.
- By 2019, bush babies dropped to about 120, and spider monkeys to about 40.
- Both populations decreased overall from 2015 to 2019, but not necessarily every year.
 - For example, the bush baby population rises slightly between 2017 and 2018.
- The population of bush babies is always greater than that of spider monkeys throughout the period.

Now check each statement:

- A. “Both decreased every year.” ■ Incorrect — bush babies didn’t fall every single year.
- B. “Bush babies always greater than spider monkeys.” ■ Correct — true for all years shown.
- C. “Spider monkeys in 2015 were 2.5 times greater than in 2018.” ■ From the graph, 2015 $\approx$ 90, 2018 $\approx$ 50, ratio $\approx$ 1.8 : 1, not 2.5 : 1.
- D. “Lowest population sizes were 32 and 44.” ■ Incorrect — lowest values are higher than these.

Final Answer

■ Option B — The population size of bush babies is always greater than that of spider monkeys.

Revision Tips

- Always read values accurately from both axes before comparing statements.
- Watch for phrases like “always” or “every year” — these are often traps.
- Describe overall trends (increase/decrease) and specific years separately.
- Graph interpretation in biology often checks your ability to distinguish trend from detail.